Last Name: First Name:

Matriculation Number:

Retake Exam WS 25/26
Artificial Intelligence 1

October 1, 2026

Please ignore the QR codes; do not write on them, they are for grading support

	To be used for grading, do not write here													
prob.	1.1	1.2	2.1	3.1	3.2	4.1	5.1	6.1	6.2	6.3	7.1	8.1	Sum	grade
total	6	4	8	12	8	6	10	8	6	7	6	9	90	
reached														

Old exams are great practice resources, but since exam styles evolve, recent exams are better models, and any course topic even if not asked before can still appear in upcoming exams. Note that these exams are for learning support only, are not exhaustive or guaranteed correct, and may not reflect the actual quiz or exam questions.

The exams here are not necessarily identical to the old exams. For technical reasons, the points may change and some sub-problems that were elided in teh exam may appear here.

In the courses Artificial Intelligence I/II and SMAI we award bonus points for the first student who reports a factual error in an old exam. (Please report spelling/formatting errors as well.)

1 General Aspects

Problem 1.1 (EU AI Act)

1. Name and briefly explain 4 requirements for high-risk AI systems stipulated by the EU AI Act. 4 pt

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Solution: The requirements include

- risk management: risks continually monitored
- data governance: transparent bias reduction in training data
- technical documentation: must be available before sold
- record-keeping: comprehensive to ensure traceability of operations
- transparency: provide all relevant information to deployers about the system
- human oversight: informed decisions on interpretation of and whether to use the results
- accuracy, robustness, security: must ensure the system works as advertised

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2. There is one risk category above high-risk. Name that risk category, and explain (in about one sentence) the regulations for using systems of this category. 2 pt

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Solution: This is unacceptable-risk. The systems are banned.

Problem 1.2 (Model-Based Agents)

A model-based agent internally keeps a variable that models the state of the environment and agent.

1. In about 2 sentences and including an example, explain how the transition model changes the value of that variable. 2 pt

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Solution: Given the current state and an action performed by the agent, it determines the subsequent state. For example, if a robot takes the action of moving left, it will update the coordinates of its position accordingly.

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2. In about 2 sentences and including an example, explain how the sensor model changes the value of that variable. 2 pt

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Solution: Given the current state and a percept, it corrects the value of the current state. For example, if the sensors show that the robot did not actually move, it updates the coordinates of its position back to the perceived value.

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2 Prolog

Problem 2.1 (Interpreting Programs)

Consider the following Prolog program for propositional formulas (without implication):

```
isProp(pv(_)).
isProp(neg(A))    :- isProp(A).
isProp(conj(A,B)) :- isProp(A), isProp(B).
isProp(disj(A,B)) :- isProp(A), isProp(B).
```

```
foo(false, pv(P), pv(P)).
foo(true,  pv(P), neg(pv(P))).
foo(false, neg(A), R)             :- foo(true,  A, R).
foo(true,  neg(A), R)             :- foo(false, A, R).
foo(false, conj(A,B), conj(R,S)) :- foo(false, A, R), foo(false, B, S).
foo(true,  conj(A,B), disj(R,S)) :- foo(true,  A, R), foo(true,  B, S).
foo(false, disj(A,B), disj(R,S)) :- foo(false, A, R), foo(false, B, S).
foo(true,  disj(A,B), conj(R,S)) :- foo(true,  A, R), foo(true,  B, S).
```

1. Give the result of the query `foo(false, neg(conj(pv("p"), neg(pv("q")))), X).` 3 pt

   Solution: `X = disj(neg(pv("p")),pv("q"))`

2. The function `foo(false,` F`,` X`)` implements a transformation that turns F into an equivalent formula X. What kind of formulas are possible results for X? 3 pt

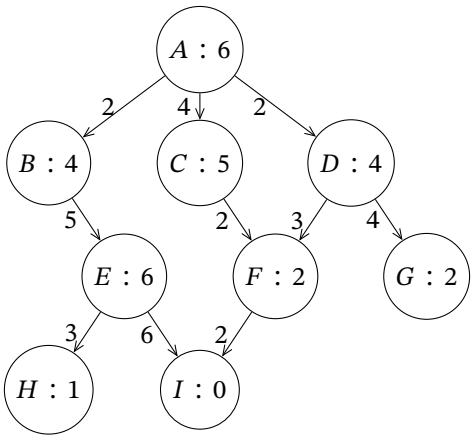
   Solution: X has negations only directly in front of propositional variables.

3. Argue (in about 2 sentences) why the query `foo(false,`f`,X)` will always find a value for `X` if `isProp(`f`)` succeeds. . 2 pt

   Solution: The Prolog program has one clause for every allowed combination of cases for the first two arguments (progress), and each case has recursive calls only for subexpressions of f (termination).

3 Search

Problem 3.1 (Search Algorithms)
Consider the graph below. Every edge is labeled with its cost.
Each node is labeled with $n : h(n)$ where n is its name and $h(n)$ its value for a heuristic h. I is the only goal.



1. Give the value $h^*(A)$. 1 pt

   Solution: 7

2. Give the node n that makes h not admissible. 1 pt

   Solution: C (because $h^*(C) = 4 < 5 = h(C)$ in violation of the definition of admissible)

3. Explain which checks you have to perform to determine if h is consistent. (You do not have to perform the checks if your explanation is precise enough.) 2 pt

   Solution: For every edge from m to n with cost c, we have to check $h(m) - h(n) \leq c$. (This check fails for $C \to F$, where the heuristic drops by 3 for a cost of 2, and $E \to H$, where the heuristic drops by 5 for a cost of 3. The edge $B \to E$ is weird because the heuristic increases even though we are getting closer to the goal, but that is irrelevant here.)

Assume you have already expanded the root node A. In each problem below, list the **next** 4 nodes that will be expanded using the respective search algorithm and (if needed) heuristic h.
If there is a tie, break it using alphabetical order.

4. depth-first search 1 pt

   Solution: B, E, H, I

5. breadth-first search 1 pt

   Solution: B, C, D, E

6. uniform-cost search 2 pt

   Solution: B, D, C, F

7. greedy search 2 pt

   Solution: B, D, F, I

8. A^*-search 2 pt

   Solution: B, D, F, I

Problem 3.2 (Abstract Search Problems)
Consider the following search problem:

- set S of states: $\{0, 1, 2, 3, 4, 5, 6, 7, 8\}$
- set A of actions: $\{0, 7, 8\}$
- transition model: $T(s, a) = \{s - a + 3\} \cap S$
- initial states $I = \{0\}$
- goal states $G = \{4\}$

1. Give an example of a state s and an action a such that a is not applicable in s. 1 pt

   Solution: Whenever $T(s, a) = \emptyset$, i.e., when $s - a + 3 \notin S$.

2. Give the state(s) that can be reached by applying the action sequence $0, 0, 7$ in an initial state. 2 pt

   Solution: The only possible state is 2.

3. Give all 3 solutions whose length is at most 8. 3 pt

   Solution: The solutions are

   - $0 \overset{0}{\to} 3 \overset{0}{\to} 6 \overset{8}{\to} 1 \overset{0}{\to} 4$
   - $0 \overset{0}{\to} 3 \overset{0}{\to} 6 \overset{7}{\to} 2 \overset{0}{\to} 5 \overset{0}{\to} 8 \overset{7}{\to} 4$
   - $0 \overset{0}{\to} 3 \overset{0}{\to} 6 \overset{7}{\to} 2 \overset{0}{\to} 5 \overset{7}{\to} 1 \overset{0}{\to} 4$

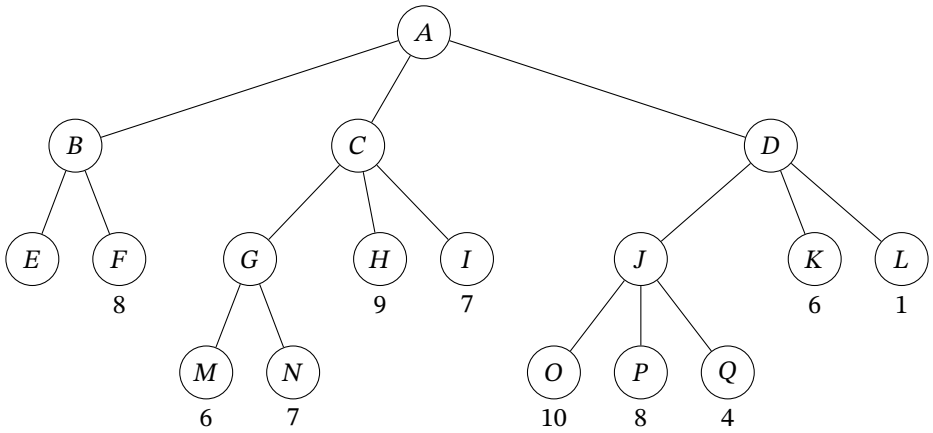
   The list of actions is the solution. The states are only included above for illustration.

4. Now assume that $I = \{0, 1\}$. You have at your disposal a black-box tool for solving search problems, but it can only process deterministic and fully observable search problems.
   Explain (in about 2 sentences) if/how you can still apply that tool in that case. 2 pt

   Solution: The problem is no longer fully observable and thus no longer deterministic. But we can run the black-box tool twice, once with $I = \{0\}$ and once with $I = \{1\}$ and take the union of the found solutions.

4 Adversarial Search

Problem 4.1 (Minimax)
Consider the following minimax game tree for the **maximizing** player's turn. The values at the leaves are the static evaluation function values of those states; some of those values are currently missing.



1. Assume E is labeled with 10. Label the node A with its minimax value. 1 pt

Solution: 8

2. Assume E is labeled with 10. Which move would be chosen by the player? 1 pt

Solution: B

3. Assume E is labeled with 10. Which nodes does $\alpha\beta$-pruning prune? We expand child nodes in alphabetical order. 2 pt

Solution: H, I, L

4. Give every possible label for the node E that would result in the player definitely choosing move B (no matter how ties are broken). 2 pt

Solution: every label with $8 \leq E$

5 Constraint Satisfaction

Problem 5.1 (Solving)
Consider the following constraint network $\langle V, D, C \rangle$:

- Variables $V = \{a, b, c, d\}$
- Domains $D_a = \{0, 1\}, D_b = \{0, 1, 2, 3\}, D_c = \{0, 1, 2, 3\}, D_d = \{0, 1, 2, 3, 4, 5, 6\}$
- Constraints C:
  - $(a = 1) \Rightarrow (b \leq 3)$
  - $(c < 2) \Rightarrow (a < 1)$
  - $b + c < 5$
  - $d < b$
  - $d = c + 2$

1. Check those boxes where one variable is arc-consistent relative to the other one: 2 pt
   - ☐ b relative to d
   - ☐ d relative to b
   - ☑ c relative to d
   - ☐ d relative to c
   - ☐ none of the above

   Solution: marked above

2. Give all solutions. 3 pt

   Solution: The only solution is $a = 0, b = 3, c = 0, d = 2$.

3. Give an inconsistent total variable assignment. 1 pt

   Solution: Any assignment that is not a solution, e.g., $a = 0, b = 0, c = 0, d = 0$.

4. Assume a CSP algorithm assigns 2 to c. Give the resulting domains of the other variables. 2 pt

   Solution: The updated domains are: $D_a = \{0, 1\}$, $D_b = \{0, 1, 2\}$, $D_d = \{4\}$.

5. Explain (in about 2 sentences) why a CSP solver can iterate forward-checking after the forward-checking step from the previous question, and how it helps here? 2 pt

   Solution: It can use $D_d = \{4\}$ to assign $d = 4$ and apply forward-checking again. That already shows $D_b = \emptyset$ and thus invalidates the assignment to c.

6 Logic

Problem 6.1 (Propositional Logic)
Consider the set $\{P, Q, R\}$ of propositional variables and the formula

$$A \quad := \quad (P \wedge Q) \Rightarrow (Q \wedge R)$$

1. Give an assignment φ that **falsifies** A. 2 pt

   Solution: Only one option: $\varphi(P) = \mathsf{T}$, $\varphi(Q) = \mathsf{T}$, and $\varphi(R) = \mathsf{F}$.

2. Create a saturated tableau for A, and explain how you can use it to obtain all falsifying assignments for A. 3 pt

Solution: The left branch below closes on Q. The right branch is open and allows reading off the only falsifying assignment.

$$\dfrac{\dfrac{A^{\mathsf{F}}}{(P \wedge Q)^{\mathsf{T}}}}{\begin{array}{l}(Q \wedge R)^{\mathsf{F}}\\ P^{\mathsf{T}}\\ Q^{\mathsf{T}}\end{array}}\qquad \begin{array}{ll} Q^{\mathsf{F}} & R^{\mathsf{F}}\end{array}$$

3. For each of the following formulas, check which of the three properties they have. (You will not be penalized for wrong answers.) 3 pt

formula	not falsifiable	satisfiable and falsifiable	not satisfiable
$A \vee \neg R$			
$R \Rightarrow A$			
$A \Rightarrow R$			

Solution: They are, respectively, not falsifiable, not falsifiable, satisfiable and falsifiable. Wrong answers were counted in the same way as non-answers. Multiple checkmarks in one row counted as a wrong answer.

Problem 6.2 (First-Order Logic)
Consider the formula A of first-order logic with equality given by

$$(\forall x.\exists y.p(x,y)) \Rightarrow (\forall x.p(x,f(x)))$$

over the smallest signature that makes it well-formed.

1. Give that signature. 2 pt

Solution: The signature with a unary function constant f and a binary predicate constant p.

2. Give a model that satisfies A. 2 pt

Solution: Lots of options. The easiest way is to make p false everywhere or true everywhere, e.g., $\langle \mathcal{D}, \mathcal{I} \rangle$ with universe $\mathcal{D} = \{0\}$, $\mathcal{I}(f)(u) = 0$ and $\mathcal{I}(p) = \emptyset$.

3. Now assume $A' := \forall x.\exists y.p(x,y)$ is the only axiom. Give a second axiom such that the models $\langle \mathcal{D}, \mathcal{I} \rangle$ that satisfy both axioms are exactly the ones where $\mathcal{I}(p)$ (which is a binary relation on $\mathcal{D}$) also represents a unary function from $\mathcal{D}$ to $\mathcal{D}$. 2 pt

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Solution: A' already ensures that $\mathcal{I}(p)$ is a total relation. So we need to additionally make it functional by requiring, e.g., $\forall x.\forall y.\forall y'.p(x,y) \wedge p(x,y') \Rightarrow y = y'$.

Problem 6.3 (Logical Systems)
Let $V = \{X, Y, Z\}$ be a fixed set of names.
Consider the logical system $\langle \mathcal{L}, \mathcal{M}, \models \rangle$ given by

- $\mathcal{L}$ is the set of formulas of the form
  - P for $P \in V$
  - $\triangle$
  - $A \oplus B$ for $A, B \in \mathcal{L}$
- The models $M \in \mathcal{M}$ are the functions $M : V \to \mathbb{N}$.
- $\models$ is defined by $M \models F$ iff $\overline{M}(F) > 0$ where $\overline{M}(F)$ is defined by
  - $\overline{M}(P) = M(P)$
  - $\overline{M}(\triangle) = 0$
  - $\overline{M}(A \oplus B) = \overline{M}(A) + \overline{M}(B)$

1. Consider the model M given by $M(X) = 0$, $M(Y) = 1$, $M(Z) = 0$. Give a formula A such that $M \models A$. 2 pt

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Solution: E.g., Y. More generally, any formula in which Y occurs. (Note that $\triangle$ behaves like falsity, and $\oplus$ like disjunction.)

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2. Give all unsatisfiable formulas. 2 pt

Solution: All formulas that use only $\triangle$ and $\oplus$. (Every formula that mentions one of the names P can be satisfied by putting $M(P) = 1$.)

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3. Consider the relation $\vdash$ defined by 3 pt

$$\mathcal{H} \vdash A \quad \text{iff} \quad A \in \mathcal{H} \quad \text{or} \quad A = A_1 \oplus A_2 \text{ for } A_1, A_2 \in \mathcal{H}.$$

Name the three properties of derivation relations. For each, say whether it holds for $\vdash$.

Solution: $\vdash$ is proof-reflexive and monotonic. (That is easy to see.) But it is not proof-transitive. (If $\mathcal{H} = \{P\}$ and abbreviating $B := P \oplus P$, we have $\mathcal{H} \vdash B$ and $\mathcal{H} \cup \{B\} \vdash B \oplus B$ but not $\mathcal{H} \vdash B \oplus B$.) Arguments were not required.

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7 Knowledge Representation

Problem 7.1 (ALC)
Consider the following description logic signature

- concept symbols: i (for instructor), s (for student), c (for course),
- role symbol m (for is-member-of) used for
  - instructors giving a course
  - students taking a course

We use an extension of $\mathcal{ALC}$, in which there are dual roles: there is a role m^{-1} that captures the relation has-as-member, e.g., $MK\ m\ AI$ iff $AI\ m^{-1}\ MK$.

1. For the signature above, give a concept axiom that captures that instructors can only be members of courses. 2 pt

   Solution: $i \sqsubseteq (\forall m.c)$

2. Give a concept axiom for the above signature that captures: courses that are taken by a student must be given by an instructor. 2 pt

   Solution: $c \sqcap (\exists m^{-1}.s) \sqsubseteq (\exists m^{-1}.i)$

3. Extend the translation from $\mathcal{ALC}$ to first-order logic with the definition of $\overline{\forall r^{-1}.C}^{fo(x)}$ for an arbitrary role symbol r (for which a predicate symbol of the same name exists in first-order logic) and an arbitrary concept C. 2 pt

   Solution: $\overline{\forall r^{-1}.C}^{fo(x)} = \forall y.r(y,x) \Rightarrow \overline{C}^{fo(y)}$

8 Planning

Problem 8.1 (Planning as Search)
Consider a STRIPS-style planning problem $\mathcal{L}$ given by

- a set P of propositions,
- a set A of triples $\langle p, a, d\rangle$ for $p, a, d \subseteq P$ (**p**reconditions, **a**dd-list, **d**elete-list),
- a set $I \subseteq P$ of propositions that are initially true,
- two sets $G^{\mathsf{T}}, G^{\mathsf{F}} \subseteq P$ of propositions that the plan is supposed to make true or false, respectively.

We want to define an equivalent search problem $\mathcal{E}$ and begin as follows:

- the states of $\mathcal{E}$ are the subsets of P,
- the actions of $\mathcal{E}$ are the elements of A,
- the single initial state is I.

1. Which states should be the goal states of $\mathcal{E}$? 3 pt

   Solution: All states $s \subseteq P$ such that $G^+ \subseteq s$ and $s \cap G^- = \emptyset$.

2. $\mathcal{E}$ is deterministic, i.e., its transition model is a partial function. Given $s \subseteq P$ and $\langle p, a, d\rangle \in A$, what is the right definition of $T(s, \langle p, a, d\rangle)$? 2 pt

   Solution: if $p \subseteq s$: $s \cup a \setminus d$; undefined otherwise.

3. $h^*_{\mathcal{L}}(s)$ for $\mathcal{L}$ is defined as the shortest plan starting in s. What is the relationship between $h^*_{\mathcal{L}}$ and the goal distance function $h^*_{\mathcal{E}}$ of $\mathcal{E}$? 2 pt

Solution: They are the same.

4. Now consider the delete-relaxation of $\mathcal{L}$. How do we have to change the definition of $T(s, \langle p, a, d\rangle)$ to obtain the corresponding search problem? 2 pt

Solution: Remove the $\backslash d$ part, i.e., if $p \subseteq s$: $s \cup a$; undefined otherwise.